

Professional Summary

Software engineer with 5+ years of experience across enterprise platforms, consultancy delivery, and early-stage product development. Currently building an **AI-powered regulatory compliance platform at Ketchup AI**, owning all frontend architecture and delivering presales demos to **Tier-1 Singapore banks**. Previously at **ByteDance**, I architected aPaaS solutions serving **180,000+ global employees**, cutting development costs by **67%** through distributed caching and reusable low-code frameworks. At **Pixium Digital**, I led full-stack projects in Elixir, React, and PHP, boosting dashboard performance by **62.5%** and accelerating deployments by **80%** with Terraform and Docker. Certified in **Data Science and Machine Learning from MIT-IDSS** and currently pursuing a **part-time BIT degree at NUS**.

Experience

Full-Stack Software Engineer — Ketchup AI

October 2025 - Present

- Owned all frontend architecture and delivery for an AI-powered regulatory compliance platform, shipping a **Next.js PWA** from zero to production. Product quality secured presales pilots with **Tier-1 Singapore banks and a regional insurer**
- Built **AI voice-to-form auto-fill** for multi-page financial compliance forms. Advisors record client conversations, AI agents extract and populate structured data in real time, replacing hours of manual data entry
- Designed a **compliance detection engine** with sentence-level traceability to audio transcripts, enabling automated **mis-selling detection** with severity triage and batch rectification workflows
- Shipped **multi-region i18n** across Singapore, Taiwan, and Malaysia with locale propagation to backend AI agents, **enterprise OAuth** (Microsoft, Google, LinkedIn, Apple with PKCE), and **PWA offline support** via custom Service Worker strategy

Software Engineer, Project Manager — ByteDance

December 2023 - July 2025

- Architected and delivered large-scale enterprise solutions on ByteDance’s internal Feishu App Engine using **Node.js, Go, and Java**, supporting mission-critical HR, finance, insurance, and legal applications for over **180,000 global employees**, achieving nearly **67% cost reduction** vs traditional development and reducing manual processes by **600-800 man-hours monthly per project**
- Drove requirements analysis and system design, acting as the key liaison between business stakeholders and technical teams, translating complex business requirements into scalable software solutions (HCM, legal workflows, HR automation), achieving user adoption rates consistently above **75%**
- Designed and deployed **high-performance backend services** using **distributed architecture, Message Queues, and Redis-based caching**, resulting in latency reductions of over **50%** and system uptimes of **99.95%** for large-scale transactional operations
- Pioneered reusable low-code modules and frameworks, enabling rapid feature rollouts and integrations, reducing project development timelines by approximately **40-50%**
- Led internal training and knowledge-sharing initiatives, mentoring junior engineers on software engineering practices and technical architecture, improving project delivery speed by up to **30%**

Full-Stack Software Engineer — Pixium Digital

March 2021 - December 2023

- Led full-stack projects in Elixir/Phoenix, JavaScript/TypeScript/React, and PHP/Yii2, achieving **100% on-time delivery** across multiple client engagements
- Automated deployments using Terraform, accelerating environment setup and releases by **80%**
- Managed **Docker-based PHP/Nginx/PostgreSQL environments**, enhancing stability and reducing downtime by **70%**
- Optimised MySQL/PostgreSQL queries, boosting analytics dashboard performance by **62.5%**
- Mentored junior engineers on coding practices and architecture, improving code quality and reducing bugs by nearly **30%**

Full Stack Software Engineer Intern — Pixium Digital

September 2020 - February 2021

- Built **web and mobile applications** in **PHP** and **Flutter** with full API integration, delivering solutions that surpassed client expectations by **20%**
- Owned **UAT and API testing workflows** across projects, cutting post-deployment defects by over **40%** while automating **Docker-based site deployments** to reduce setup effort by **50%**

Education

Bachelor of Information Technology — National University of Singapore

July 2025 - Present

- Pursuing a **part-time BIT degree** to deepen technical and architectural expertise

Diploma in Information Technology — Singapore Polytechnic

April 2018 - May 2021

- Specialised in **UI/UX Design** and **Data Structures & Algorithms**, building strong foundations for software engineering
- Delivered a **Java-based full-stack final-year project** for Sentosa Development Corporation (3rd place), in a collaborative five-member team

Certifications and Awards

Data Science and Machine Learning: Making Data-Driven Decisions - MIT Schwarzman College of Computing

January 2024

Skills

- Languages: JavaScript, TypeScript, Python, Go, Elixir, Java, PHP, Dart, Bash
- Frameworks & Libraries: Next.js, React, Node.js, Phoenix, LiveView, Flutter, Tailwind CSS, Vue
- Infrastructure & Databases: AWS, Terraform, Docker, Redis, PostgreSQL, MySQL, Firebase, Nginx, Grafana
- Testing & Tools: Vitest, Playwright, Jest, MSW, Postman, Figma